Report Title:

Statistical Analysis and Forecasting of Monthly Precipitation of Adelaide Airport between 1990 and 2025.

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Aim:

This paper will examine the rainfall patterns over a long period (1990-2025) at the Adelaide Airport by studying the past records of precipitation over these years. Rainfall variation plays a vital role in engineering works, such as flood management, stormwater drainage systems, and infrastructure. But because of natural climatic changes, it may be difficult to see any regular patterns and trends in precipitation.
This paper aims to establish whether the rainfall amounts have increased, decreased, or maintained the same level over time, study how the rainfall is distributed annually, and the degree of fluctuation of rainfall across time. The research will be conducted to determine the unusual rainfall events, which can signal dangerous weather conditions, and create a time-series forecasting model to predict future changes in rainfall. The main purpose is to offer evidence-based information that will help in planning and engineering decision-making in the environment.

Research Focus:

The study is aimed at assessing the mean and range of rainfall in Adelaide Airport monthly over 35 years between the years 1990 and 2025. The initial goal is to examine the changes in rainfall every month and to determine trends regarding the distribution, mean, and maximum values of rainfall. It is a method that looks at the whole range of rainfall information as opposed to the average. This helps in understanding how monthly rainfall changes or remains the same.
A box plot may be applied in order to compare the level of precipitation monthly. This allows watching of the median precipitation, interquartile ranges, and anomalies. This will allow us to distinguish the months where the rainfall occurrence is unpredictable or unforeseeable and those where the patterns are more predictable and steady. The hypothesis of the experiment is to find out whether there is an annual pattern in rainfall, that is, specific months may have higher or lower precipitation levels.
This is unlike other group studies, like bar charts and scatter plots, in that it analyzes variability and distribution but not averages or relationships of variables. This gives us more understanding of the behavior of human beings in several seasons and seasons where there is a lot of rain.

Data Set:

The data used in this paper were sourced from the official climate data repository of the Bureau of Meteorology. The Adelaide Airport Meteorology Office (Station No. 023034) was chosen because it is reliable, has long-term records, and minimal omissions in data.
The data is monthly rainfall data between the years 1990 and 2025, with a total period of 35 years. As monthly data was chosen instead of daily data, to reduce short-term noise and fluctuations, it will be easier to identify long-term trends and patterns of a season. It helps to analyze the statistics and minimise the problem of missing data.
The dataset was pre-filtered for analysis by removing extraneous variables and keeping Year, Month, and Rainfall. We used only quality-controlled observations, and this data was legitimate. The data was neither missing nor negative. We retained the figures of zero rainfall because it represents the lack of precipitation and is necessary to analyze the seasonal patterns and variability. The resultant data can be taken as clean and reliable, and can be analyzed statistically.

Preliminary Plot:

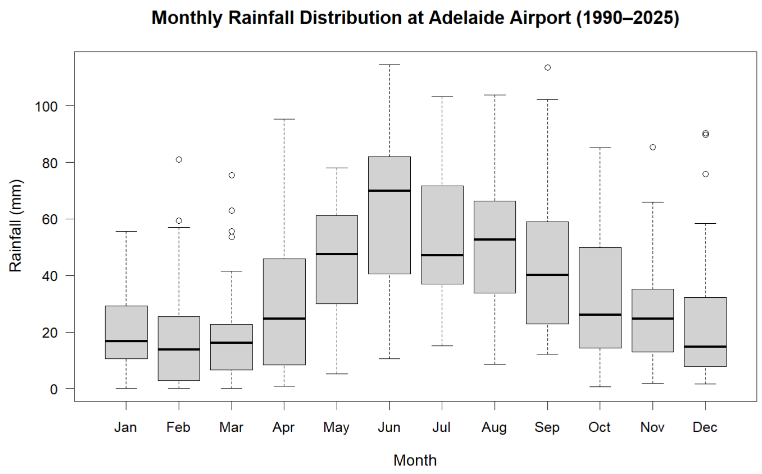

Figure 1: Boxplot of monthly rainfall distribution at Adelaide Airport (1990–2025)

The boxplot shows the monthly rainfall at Adelaide Airport between 1990 and 2025. Each box illustrates the median, the interquartile range, and the potential outliers. This facilitates the comparison of monthly differences in rainfall in detail.

There is a tendency for annual precipitation to be seasonal. The median rainfall and the range of values are higher in the winter months, especially in the months of June, July, and August. It means that there are months that are more precipitated and more variable. Conversely, during the summer months, the mean of January, February, and December decreases and becomes less fluctuating, which means that the weather is drier and more predictable.

Some months have outliers larger than the upper whiskers of the data, indicating that there are instances of extremely high quantities of rainfall. This is mainly noticeable in transitional months and winter when there is a lot of rainfall, and it is intermittent. Very large or very small numbers are noteworthy since they can reflect the times of high risk of flooding or when drainage systems are overloaded.

The boxplot shows how rainfall in Adelaide Airport fluctuates on a seasonal basis. The analysis of the research problem is in the sense that it shows the monthly changes in rainfall and identifies the period of higher variability and intense periods of rainfall occurrences.

Conclusion:

This is a preliminary research addressing the monthly rainfall at Adelaide Airport between the years 1990 and 2025 and the temporal variations. The boxplot also showed clearly that precipitation is not distributed evenly throughout the year, but it has particular seasonal distributions. The winters were more rainy and diverse, and the summers drier and more predictable.

The fact that the outliers are observed in different months means that there are indeed large quantities of rainfall, and it is significant to look at the extremes and variation rather than just the averages. These findings are helpful to the topic of research as they indicate the variations in the rainfall pattern per month and the months that are usually the most variable.

The research gives good ground to any additional statistical investigation, such as trend, variability, and time-series predictions. The dynamics of rainfall patterns are important to the engineering activities of designing drainage systems, flood management, and the development of infrastructure.